BT503 Solved By: Muhammad Nadeem 24/02/20

Current Final Term MCQs

1. Reintroduction of these **"trained"** bacteria to the polluted sites should give them an advantage over the indigenous bacteria as they would be better suited to survive and remediate the contamination.
2. The DNA construct contains **regions of DNA** which are complementary to the plant DNA to enable the inserted piece to recombine into the plant genome.
3. Attack by insects not only causes damage to the plant but also provides a route **for bacterial or fungal infection.**
4. Bacteria which Thrive at temperatures above 45°C is called as **Thermophiles**
5. Bacteria which Thrive at temperatures above ~85°C. is called as **Hyper thermophiles**
6. Following factor in linked to toxicity of pollutants…..??????
7. Which attribute of pollutant is not linked to bio magnification **long-lived, • mobile, • soluble in fats, • biologically active other than these option**
8. Following substances cannot use to make the vessel of bio filter **concrete, metal or durable plastic,(other than these)**
9. Following cannot be adopted to increase the efficiency of phosphorus removal by activated sludge method **?????**
10. Following is true about imhoff tank used in anaerobic filter system **(see page no 138 midterm)**
11. When facultative ponds receive raw sewage, they are also called **primary** ponds.
12. In general, enzymatic biosensors are based on **the selective inhibition of specific** enzymes by different classes of compounds.
13. Which enzyme are used in leather industry **Traditionally, pancreatic enzymes were employed**
14. Explosives of characterized by presence of _____group??????
15. Which statement about chlorinated hydrocarbons is correct **fixation of carbon by phytoplankton.**
16. Main use of DDT other than fighting crop insects effective in controlling the Anopheles mosquito **malaria**
17. The term "endophytes" is usually used to denote **non-pathogenic** microorganisms.
18. The steps of solid waste management are listed below what is correct **(see the final term ppts page no 396…)**
19. Statement about on-sito disposal of waste is incorrect

Advantages: • Bins are potentially a highly hygienic and sanitary management method; and • Final disposal of waste is well away from dwelling areas. Constraints: • Collection, transportation and human resources are required; • System takes time to implement; and • Efficient management is essential. (Other than these)

20. Statement about phytoremediation is false **(see the final term ppts page no 392…)**
21. State about sampling for situ monitoring is true **(see the final term ppts page no 182…)**
22. Following property of hydrocarbon does not contribute poor biodegradation **(see the final term ppts page no 708…)**
23. In planning studies for the implementation of the wastewater treatment, the following points must be clearly addressed: **• Environmental impact studies on the receiving body • Treatment objectives • Treatment level and removal efficiencies**

دنیا میں سب سے مشکل کام اپنی اصلاح اور سب سے آسان کام دوسروں پر نکتہ چینی کرنا ہے

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

24. Toxicity represents the potential damage to life and can be both short and long term. It is related to
• Type of the pollutant, • Concentration,• The time of
25. In order for bio magnification to occur, the pollutant must be:**• Long-lived,• Mobile,• Soluble in fats,• Biologically active.**
26. bio filters can reduce odour release by **95% or more**
27. For higher efficiencies of phosphorus removal, effluent polishing methods can adopted: **Addition of coagulants**
28. In the Inhofe tank, settling occurs in the…… **upper compartment**
29. In the soil, a pollutant has basically four possible destinations:
• Retention in the soil matrix• Retention by the plants• Appearance in the underground water
• Collection by Under drains
30. Facultative ponds which receive raw sewage is also known as ……..**primary ponds**
31. The bio scrubber is not itself truly a biological treatment system, but rather a highly efficient method of removing odour components by dissolving them.
32. . In the Imhoff tank, settling occurs in the upper compartment **(settling compartment).**
33. . Biodiesel is derived mostly from---- **vegetable oil**
34. . DDT was of great benefit to humans in their fight against------- **malaria**
35. . Two characteristics of chlorinated pesticides that led to great harm are
• Their resistance to biodegradation and• their hydrophobicity
36. . A characteristic of explosives is the presence of ----------groups. **Nitro (-NO2)**
37. . Endophyte is micro pathogen
38. . up flow anaerobic sludge blanket reactor stand for **(UASB)**
39. Water treatment accounts for **25%** of the total global environmental market.
40. Basic equipment required in Preliminary treatment involves screen, **grit chamber and flow meter**
41. UASB stands for ----------**up flow anaerobic sludge blanket reactor**
42. Important factors to be considered when evaluating and selecting unit operations and processes except:
a) Process applicability, b) Applicable flow, c) Influent characteristics, **d) Sterilization of vessels**
43. What is false about final disposal of sludge in ocean?
a) After pre-conditioning, the sewage is disposed in the sea **b) It has beneficial uses**. c) It gives pollution to oceans
44. Which statement is true about Site monitoring analysis sampling?
a) Sampling involves two distinct activities b) Does not define the objectives of the program
c) Characterize the samples in terms of description, location, batch number, size,
45. **Rhizodegradation** refers to the biodegradation of contaminants in the soil by edaphic microbes enhanced by the inherent character of the rhizosphere itself.
46. The correct sequence of Initial Steps in Solid Waste Management:
1. Identify the types of waste. 2. Identify the sources of waste. 3. Determine the potential health hazards from waste. 4. Determine the volume of waste generated 5. Identify safe collection method/s. 6. Identify safe transportation method/s. 7. Identify safe disposal method/s. All of these.
47. What is false about Collection stage?
a) Simply refers to how waste is collected for transportation to the final disposal site.
b) Should be carefully planned to ensure that storage facilities do not become overloaded.
c) Collection intervals and volumes of collected waste must be estimated.



ایماندار کو غصہ دیر سے آتا ہے اور جلدی دور ہو جاتا ہے

d) This is the stage when solid waste is transported to the final disposal site

48. False about On-site disposal?
    a) waste volumes are small, b) plenty of space is available, and c) waste is largely organic or recyclable.(biodegradable)
    d) waste volumes are large
49. What is true about Biodiesel? **It is derived mostly from vegetable oils**
50. The term "endophytes" is usually used to denote **non-pathogenic microorganisms.**
51. A tumor-like growth, seen as a crown gall in plants. **Agrobacterium tumefaciens**
52. What is false about dioxins?
53. a) Dioxins released as by-products in a number of chemical processes
54. b) The manufacture of PVC plastics c) carcinogenic **d) Do not pose threats to environment**
55. Explosives use **nitro** functional group
56. Aerobic Co-metabolism of **CAHs-II requires** NADH to provide the 2H
57. What is false about poor biodegradation of hydrocarbons?
    a) Complexity of the molecule b) Branched or aromatic rings are difficult to degrade c) Low water solubility.
    d) Easily degraded by enzymatic attack
58. Nonspecific enzymes are **oxygenases** that initiate the oxidation of hydrocarbons such as methane or toluene.
59. Western blotting is used to detect **proteins**
60. Due to continuous arrival of food in the form of BOD to the aeration tank, bacteria grow and reproduce continuously. If an indefinite population growth were allowed…-------following problems arised except
    a) The secondary sedimentation tank would become overloaded,
    b) The solids would not settle well
    c) They would start to leave with the final effluent…
    d) option d will be correct option
61. Wastewater is applied to the soil, supplying water and nutrients necessary for plant growth. Which is not true?
    a) Some liquid evaporates, b) Percolates into the soil, c) Absorbed by the plants… **d) Generation of aerosols**
62. Which statement is not true regarding desulphurization of oil and coal?
    a) washing pulverized coal and the use of fluidized bed technology b) Both organic and inorganic sulphur is present in coal c) Aerobic, acidophilic chemolithotrophs like Thiobacillus species, for desulphurization of the inorganic sulphur in coal. **d) Option d will be correct**
63. The metals release in air by all following process except
    a) Processes of weathering, b) Erosion c) Volcanic activity **d) Earthquake**
64. mer A gene codes for the enzyme, **mercuric ion reductase**

زندگی میں کامیابی کا یہی راز ہے کہ پریشانیوں سے پریشان مت بنو

جھوٹ انسان اور ایمان دونوں کا دشمن ہے

65. What is true regarding zooplankton? **Rely on tides and currents as a transport**
66. Facultative ponds which receive raw sewage is also known as **primary ponds**
67. In the soil, a pollutant has all the possible destinations except?
    a) Retention in the soil matrix b) retention by the plants c) appearance in the underground water
    d) Option d will be correct
68. What is not true about bio scrubbers?
    a) Removing odour components by dissolving them. b) Most appropriate for hydrophilic compounds
    c) Remove toxic metals
69. One such recombinant is a plant where the fatty acid composition in the seed has been modified to produce triacylglycerol's containing elevated levels of trierucinic acid suitable for use in the polymer industry. **Arabidopsis thaliana**
70. Unlike other marines species, sharks are characteristically untroubled by **bio fouling** and studies of their skin have established that its unique texture coupled with its inherent antimicrobial properties help keep them clean of fouling organisms.
71.) Factors related to bio magnifications except
    a) Mobile/transfer b) solubility in fats, c) biologically active **d) source**



Mid Term Mcqs

1. Natural estrogens deactivated by..... **Glucuronidation**
2. Insect virus that causes over expression of genes in molecular biology .....**baculo virus**
3. . Acute toxins .....**All**
4. By using recombinant technologies scientists are now producing.....**Insulin**

5.are combustible materials, containing within themselves all oxygen needful for their combustion, which burn but do not explode, and function by producing gas which produces an explosion…..**Propellants or low explosives**

6.are types of waste stabilization pond used for biological treatment of industrial and domestic wastewater .....**Facultative lagoon**

7. PAH-containing fractions, by animal test systems ..... evidence have been obtained that PAH containing..... **(skin painting, lung implantation)**

8. Pentane is utilized as a ……in the preparation of phenol .....**foaming**
9. Polystyrene, as a propellant for aerosol sprays, as a filling of low-temperature **thermometers**, and as reference compound in gas chromatography.
10. Herbicide resistant plants..... **pests**
11. Average temperature of ocean…..**1–3°C**
12. Which of ....does not use to make vessel of biofilters.... **(may be formic sheet)**
13. Which of the following is not fate of water.....**(may be generation of aerosol)**

14. Which of the following does not classify as a substitution material ..... (other than those options) **materials is a major potential avenue for the environmental biotechnology**

15. Which of the following statement about desulphurization of oil and coal is false .... **(See the ppt page no 390 to 400)**
16. Metals are released into the environment by all of the following except **(Earthquake)**
17. When facultative pond receives raw sewage they are called ..... ponds. **(primary pond)**
18. Which of the following is not linked to biomagnifications
19. Why frogs are helpful as bioindicators ..... **Frogs are also Bioindicators; influenced by changes that take place in their freshwater and terrestrial habitats**
20. For land base application soil type best suited to biotechnological intervention are..... **(Sand and gravel)**
21. Attack by insect not only cause damage to the plant but also provide route..... **for bacterial or fungal infection**
22. Insertion of foreign DNA in the vector resulting in the loss of the future.
    (marker gene)
23. Methogenic contortion of over 100 bacterial clone with the capability to digest ...
    (terephthalate)
24. Preliminary treatment involves the removal of coarse..... **solids .....** only
25. Due to continuous arrival of food in the form of BOD to the aeration tank, bacteria grow and reproduce continuously. **If an indefinite population growth**
26. Septic tanks are also a form of ,,,,,,,,,**primary** ,,,,,,,,,treatment
27. Mere a gene is one of a .....**cluster**.....of genes involved in bacterial detoxification of mercury, and is the one coding for the enzyme, mercuric ion reeducates, which converts mercury from an ionic to a volatile form
28. Facultative ponds which receive raw sewage is also known as..... **primary ponds**
29. Facultative ponds which receive raw sewage is also known as primary ponds ..... **Retention by the plants**
30. In the Inhofe tank, settling occurs in the… ..... **upper compartment**

31. Second strategy is usually .....**plasmid**
32. Fundamental requirements are.....**all of given above**
33. Attack by insects is cause……**micrococcus lentos**
34. Na+/H+ antiport, ….**the transport of sodium and hydrogen ions** …in opposite directions across a membrane
35. Frog are bio indicator by ..... **Change in fresh and terrestrial**
36. For higher efficiencies of phosphorus removal, effluent polishing methods can adopted:
Addition of coagulants
(Metallic ions):
Phosphorus precipitation
Effluent filtration: removal of the phosphorus present in the suspended solids in the effluent
Combination of the addition of coagulants and filtration

37. Wastewater is applied to the soil, supplying water and nutrients necessary for **plant growth.**
38. Bio filter consists of a relatively large vessel or container, typically made of cast concrete, metal or **durable plastic**
39. The metals released in air by the processes of weathering, erosion and volcanic activity
40. The nature of the soil or water which harbors the pollution can have a major effect on the actual expressed end-result.
41. Zooplanktons are microscopic animals living near to the surface of the water body. They are poor swimmers, instead relying on tides and currents as a transport mechanism. They feed upon phytoplanktons, bacterioplanktons, or detritus (i.e. marine snow).
42. Phytoplanktons, also known as microalgae, are similar to terrestrial plants in that they contain chlorophyll and require daylight to live and develop.
43. Mere a gene is one of a cluster of genes involved in bacterial detoxification of mercury, and is the one coding for the enzyme, mercuric ion reeducates, which converts mercury from an ionic to a volatile form.
44. The insect virus, Baculovirus, has been shown to be the method of choice for the over expression of genes in many applications of molecular biology.
45. A strain of Pseudomonas putrid known to degrade PCBs
46. Facultative ponds which receive raw sewage is also known as primary ponds
47. Bio substitution are: **Biofuels, Polymers, Biodegradable**
48. septic tank can be a single-chamber tank or a two-compartment tank **(called an Imhoff tanks).**
49. In the Imhoff tank, settling occurs in the upper compartment **(settling compartment).**
50. In the soil, a pollutant has basically four possible destinations:

- retention in the soil matrix,
- retention by the plants,
- appearance in the underground water,
- collection by underdrains

51. Facultative ponds which receive raw sewage is also known as **primary ponds**

52. A strain of Pseudomonas putida known to **degrade PCBs**





53. Which of does not use to make vessel of biofilters.....**(may be formic sheet)**

54. Which of the following is not fate of water.....**(may be generation of aerosol)**

55. Which of the following does not classify as a substitution material.....**(other than those options) materials is a major potential avenue for the environmental biotechnology**

56. Which of the following statement about desulphurization of oil and coal is false.....**(See the ppt page no 390 to 400)**

57. Metals are released into the environment by all of the following except..... **(Earthquake)**

58. When facultative pond receive raw sewage they are called ponds.....**(Primary pond)**

59. Which of the following is not linked to biomagnifications ...**(other than those) • long-lived, • mobile, • soluble in fats, • biologically active.**

60. Why frogs are helpful as bioindicators … **Frogs are also Bioindicators; influenced by changes that take place in their freshwater and terrestrial habitats**

61. For land base application soil type best suited to biotechnological intervention are **(Sand and gravel)**

62. **The septic tank can be a single-chamber tank or a two-compartment tank (called an Inhofe tank).**

63. In the Inhofe tank, settling occurs in the upper compartment **(settling compartment).**

64. Attack by insect not only cause damage to the plant but also provide route … **for bacterial or fungal infection**

65. Insertion of foreign DNA in the vector resulting in the loss of the future..... **(marker gene)**

66. Methogenic contortion of over 100 bacterial clone with the capability to digest …**(terephthalate)**

Preliminary treatment involves the removal of coarse .....**solids only**

BSc Zoology 4th Semester

جو شخص ناکامیوں سے ڈر کر بھاگتا ہے کامیابی اُس سے ڈر کر بھاگتی ہے

عقل مند اپنے عیب خود دیکھتا ہے اور بیوقوفوں کے عیب دنیا دیکھتی ہے

Current Final Term

BT 503 Final Current Solved – Mostly Question Solved from Internet

Question and Answers

Feature of phenol that make it pollutant?

Phenols and some detergents may also mimic the activity of oestrogen.

Name enzyme equipment and solution used in genetic engineering?

the enzymes, solutions and equipment necessary to perform the procedures; the desired piece of DNA to be transferred; a cloning vector; and the recipient cell. It is also essential to have some means of determining whether or not the transfer has been successful. This is achieved by the use of marker genes.

Importance of anaerobic zone?

The anaerobic zone gives good conditions for the development or selection of a large population of phosphorus accumulating organisms.

Biological process of phospherial removal activated sludge treatment?

The **process** of **removing phosphorus** by accumulating it with biomass is referred to as enhanced **biological phosphorous removal** (EBPR) **process**. The **phosphorous removal** efficiency for **biological** systems depends on the **phosphorus** content of the **sludge** removed and the efficiency of the solids separation **process**. **(Internet)**

When the biological excess sludge is wasted from the system, phosphorus is removed… For higher efficiencies of phosphorus removal, effluent polishing methods can adopted: • addition of coagulants (metallic ions): phosphorus precipitation • effluent filtration: removal of the phosphorus present in the suspended solids in the effluent • combination of the addition of coagulants and filtration. **(PPTS)**

Photodegradation and name its forms?

Photodegradation involves the **process** of oxidation (loss of electrons, increase in oxidation state) and hydrolysis (split of chemical bonds by adding water).....**Photodegradation** does not talk about the ageing or degradation of

What are living conditions extremophile give example?

Three **examples** of **extremophiles** are Picrophilus torridus (a thermoacidophile adapted to hot, acidic **conditions**), Antarctic krill (a psychrophile), **and** the Pompeii worm (a thermophile).

What are application of biosensor in detection of biocides?

Biosensors have been extensively applied in clinical, food and environmental areas due to the advantages of fast **detection** speed, high selectivity and sensitivity (Malhotra and Chaubey, 2003). In this study, the contribution of bioluminescence and luminometer makes a **biosensor** for **detection** of water pollution

Explain genomic libraries?

A **genomic library** is a collection of the total **genomic** DNA from a single organism. The DNA is stored in a population of identical vectors, each containing a different insert of DNA. ..... The fragments are then inserted into the

Biofilter

Biofiltration is a pollution control technique using a bioreactor containing living material to capture and biologically degrade pollutants. Common uses include processing waste water, capturing harmful chemicals or silt from surface runoff, and microbiotic oxidation of contaminants in air

Biodiesel?

Biodiesel is a form of diesel fuel derived from plants or animals and consisting of long-chain fatty acid esters. It is typically made by chemically reacting lipids such as animal fat, soybean oil, or some other vegetable oil with an alcohol, producing a methyl, ethyl or propyl ester.

Plant associated with fungi?

Mycorrhizae: Associations of fungi with roots of vascular plants

mycorrhizae, mycorrhiza or mycorrhizas) is a symbiotic association between a **fungus** and a **plant**. The term mycorrhiza refers to the role of the **fungus** in the **plant's** rhizosphere, its root system. Mycorrhizae play important roles in **plant** nutrition, soil biology and soil chemistry.

Water system?

water system. a river with all its tributaries. a **system**, including reservoirs and mains, for storing and supplying **water** for use by a community.

How plant can use be reduced pollution?

Plants improve air quality through several mechanisms: they absorb carbon dioxide and release oxygen through photosynthesis, they increase humidity by transpiring water vapor through microscopic leaf pores, and they **can** passively absorb **pollutants** on the external surfaces of leaves and on the **plant** root-soil system.

How animal work as bioindicator two example?

Bioindicators are living organisms such as plants, planktons, animals, and microbes, which are utilized to screen the health of the natural ecosystem in the environment.

Example of a **bioindicator** is lichens. These plants, which live on surfaces such as trees or rocks or soil, are very sensitive to toxins in the air.

Define phytoremediation?

Phytoremediation /ˌfaɪtəʊrɪˌmiːdɪˈeɪʃən/ refers to the technologies that use living plants to clean up soil, air, and water contaminated with hazardous contaminants.

What are biodegradation?

Biodegradation is the breakdown of organic matter by microorganisms, such as bacteria and

Bacteriophage?

In bacteria, the most likely candidates for genetic transfer are plasmids and bacteriophages.

Biosensor advantage?

Advantages of tissues as **biosensors** include the following: easier to immobilize compared to cells and organelles. the higher activity and stability from maintain enzymes in natural environment.



Sources dioxin?

- **Combustion** sources, e.g. municipal waste or medical waste incinerators and private backyard barrel burning.
- Metal smelting.
- Refining and process sources.
- Chemical manufacturing sources.
- Natural sources

Method to get quality crop?

The average **quality** can be increased by planting seeds into fertilized soil. Increasing Farming Skill also increases the chance to harvest a **quality crop**. If a Seed Maker turns a higher-**quality crop** into seeds, the seeds are no more likely to produce higher-**quality crops** when grown and harvested.

Phase sludge water in sec wastewater treatment?

Sludge is a semi-solid slurry that can be produced from a range of industrial processes, from **water treatment**, **wastewater treatment** or on-site sanitation systems. ... Industrial **wastewater treatment** plants produce solids that are also referred to as **sludge**.

In which filed biotechnology used?

• manufacturing process, • waste management, • pollution control

Thermophile and its used in PCR? Thermophile and application taq polymerase in PCR?

In PCR: DNA Polymerase of Thermus aquaticus (Taq Polymerase), Pyrococcusfuriosus (Pfu Polymerase). Clean energy: bioethanol production, hydrogen production fuel cells, etc.

Define Extremophiles and name three categories?

Extremophiles are animals that live and thrive under extreme environmental conditions
Psychrophiles can tolerate these circumstances.
Halophiles… survive intensely saline environments, such as exist in natural salt lakes or salt evaporation ponds.
Acidophiles thrive in the conditions of low pH, typically below 5

Extremophile in mature?

??

Trickling filter?

A **trickling filter** is a type of wastewater treatment system. It consists of a fixed bed of rocks, coke, gravel, slag, polyurethane foam, sphagnum peat moss, ceramic, or plastic media over which sewage or other wastewater flows downward and causes a layer of microbial slime (biofilm) to grow, covering the bed of media.

Advantage and disadvantage of incineration?

Advantages: • Drastic volume reduction, • Sterilisation
Disadvantages: • High costs, • Ash disposal, • Atmospheric pollution.

Effect of dioxin on human?

Dioxins may result in skin lesions, such as chloracne and patchy darkening of the skin, and altered liver function. Long-term exposure is linked to impairment of the immune system, the developing nervous system, the endocrine system and reproductive functions

Long 5 Marks

Explain environmental factor bioremediation?

Bioremediation is the use of living organisms, primarily microorganisms, to degrade the **environmental** contaminants into less toxic forms. It uses naturally occurring bacteria and fungi or plants to degrade or detoxify substances hazardous to human health and/or the **environment**.

Biodiesel advantage?

Biodiesel is a domestically produced, clean-burning, renewable substitute for petroleum diesel. Using biodiesel as a vehicle fuel increases **energy** security, improves air quality and the environment, and provides safety benefits.

Solid waste step?

1. Identify the types of waste.
2. Identify the sources of waste.
3. Determine the potential health hazards from waste.
4. Determine the volume of waste generated
5. Identify safe collection method/s.
6. Identify safe transportation method/s.
7. Identify safe disposal method/s.

Genetic modification in popular plant used to removal mercury? Describe how?

Genetic modification of a poplar to enable mercury to be removed from the soil and converted to a form able to be released to the atmosphere.

mer A gene gene is one of a cluster of genes involved in bacterial detoxification of mercury, and is the one coding for the enzyme, mercuric ion reductase, which converts mercury from an ionic to a volatile form.

Initially the gene was transferred to Arabidopsis thaliana. The gene was transferred by gene guns to poplar tree (Liriodendron tulipifera) embryogenic material. The resulting poplar plantlets were found to exhibit tolerance to mercury and to volatalise it at 10 times the rate observed in untransformed yellow poplar plantlets.

Factor affect bioremediation?

The factors that directly impact on bioremediation are energy sources (electron donors), electron acceptors, nutrients, pH, **temperature**, and inhibitory substrates or metabolites. One of the primary distinctions between surface soils, vadose zone soils and groundwater sediments is the content of organic material.





Properties of soil which has effect on pollution?

The application of waste or **polluted** water into the **soil** alters its physical and chemical, thereby affecting the growth of agricultural crops and other living organisms. Even the extensive agricultural practices such as fertilizer and pesticide application deteriorate the **soil quality**.

Detail note on environmental biotechnology?

Environmental biotechnology. **Environmental biotechnology** is the branch of **biotechnology** that addresses **environmental** problems, such as the removal of pollution, renewable energy generation or biomass production, by exploiting biological processes.

Explain recombinant microorganism especially yeast and microorganisms?

GE of microorganisms for use in environmental biotechnology: expansion of metabolic pathways either to modify the existent metabolic capability, or to introduce new pathways.

Many examples… A strain of Escherichia coli into which was engineered some 15 genes originating from Pseudomonas. These were introduced to construct a pathway able to produce indigo for the dyeing of denim (Bialy, 1997). The traditional method requires the use of toxic chemicals.

Long 10 Marks

Briefly describe category of solid waste?

Different categories of solid waste include:

- Non-combustibles: Metal, tin cans, bottles, stones, etc.
- Ashes/dust: Residue from fires used for cooking.
- Bulky waste: Tree branches, tyres, etc.
- Dead animals: Carcasses of domestic animals and livestock
- Hazardous waste: Oil, battery acid, medical waste.
- Construction waste: Roofing, rubble, broken concrete, etc.

Genomic Library?

Genomic libraries Collection of total Genomic DNA from an organisms. The genomic DNA is isolated, purified and cut up into pieces of a size suitable to be inserted into a cloning vector, following by ligation and transformation into a host cell. These DNA pieces may either be ligated as a total mixture, into a suitable vector to produce a genomic library, or a specific piece may be isolated and prepared as described ago. Genomic libraries are very useful, as they may be amplified, and accessed almost limitlessly, to look for a specific DNA Sequence. If the genomic library is of a eukaryotic origin the genes will contain non-coding regions, called introns. (Prokaryotes do not contain introns). This is a problem if the gene is to be expressed. This problem can be avoided by using cDNA.

Sludge treatment procedure?

Sewage **sludge treatment** describes the **processes** used to manage and dispose of sewage **sludge** produced during sewage **treatment**. **Sludge** is mostly water with lesser amounts of solid material removed from liquid sewage. ...

Reducing **sludge** volume may increase the concentration of some of these toxic chemicals in the **sludge**.



Phytovolatilization?

Phytovolatilization is a process, in which plants take up contaminants from soil and release them as volatile form into the atmosphere through transpiration. The process occurs as growing plants absorb water and organic contaminants.

What is cloning factor? Why important in genetic engineering?

Cloning allows for the creation of multiple copies of **genes**, expression of **genes**, and study of specific **genes**. ...

Plasmids have been highly **engineered** as vectors for molecular **cloning** and for the subsequent large-scale production of **important** molecules, such as insulin.

Southern blotting and western blotting?

In Southern blotting, DNA fragments are resolved by electrophoresis on a gel. These resolved fragments are then shifted to a membrane (nylon or nitrocellulose).

Western blotting: Proteins (e.g., antibodies antigens interaction)

Advantages and disadvantages of incineration (2)

Incineration:

Advantages:

•Drastic volume reduction,
•Sterilization.

Disadvantages:

•High costs,
•Ash disposal,
•Atmospheric pollution.

Manipulation of bacteria without genetic engineering (3)

A general procedure is to take a sample of bacteria from, at, or near, the site of contamination from which a pure culture is obtained in the laboratory and identified, using standard microbiology techniques. The 'training' may be required either to improve the bacterium's tolerance to the pollutant or to increase the capabilities of pathways already existing in the bacterium to include the ability to degrade the pollutant, or a combination of both.

3 phases present in submerged bio filter (3)

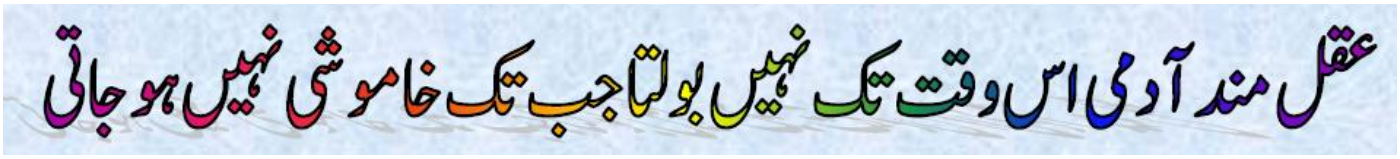
Typically, a biofilter has two or three phases, depending on the feeding strategy (percolating or submerged biofilter): a solid phase (media); a liquid phase (water); a gaseous phase (air).

Co-metabolism

Co-metabolism is defined as the simultaneous degradation of two compounds, in which the degradation of the second compound (the secondary substrate) depends on the presence of the first compound (the primary substrate).

Three forces are in the global environment market

A number of forces over which it has little or no control affect a company's marketing activities. Taken together, they make up its external marketing environment, which includes regulatory and political activity, economic conditions, competitive forces, changes in technology, and social and cultural influences.



Phytodegradation

Phytodegradation which is also known as phyto-transformation is the breakdown of contaminants taken up by plants through metabolic processes within the plant, or the breakdown of contaminants surrounding the plant through the effect of enzymes produced by the plants.

Phytoremediation

The treatments of pollutants or waste (as in contaminated soil or groundwater) by the use of green plants that remove, degrade, or stabilize the undesirable substances (such as toxic metals).

Three sources of dioxin production

According to the most recent US EPA data, the major sources of dioxins are broadly in the following types:
Combustion sources, e.g. municipal waste or medical waste incinerators and private backyard barrel burning. Metal smelting.
Refining and process sources.

Conjugation transduction plasmid difference

Conjugation is the transfer of circular DNA called plasmids through cell to cell contact. Transformation is the uptake of 'free' DNA from the environment. Transduction is the transfer of DNA by bacteria-specific viruses called bacteriophage.

Biodiesel

Biodiesel is a form of diesel fuel derived from plants or animals and consisting of long-chain fatty acid esters. It is typically made by chemically reacting lipids such as animal fat (tallow), soybean oil, or some other vegetable oil with an alcohol, producing a methyl, ethyl or propyl ester.

